Topic Four

Worms and other Malwares

- Whether wired or wireless, computer networks are essential to everyday activities.
- Individuals and organizations depend on their computers and networks for functions such as email, accounting, organization, and file management.
- Intrusion by an unauthorized person can result in costly network outages and loss of work.
- Attacks to a network can be devastating and can result in a loss of time and money due to damage or theft of important information or assets.

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- Intruders can gain access to a network through software vulnerabilities, hardware attacks, or even through less high-tech methods, such as guessing usernames and passwords.
- Intruders who gain access by modifying software or exploiting software vulnerabilities are often called threat actors.
- When the threat actor gains access to the network, four types of threat may arise:

✓Information theft

✓Data loss and manipulation

✓Identity theft

✓Disruption of service

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- Information theft is breaking into a computer to obtain confidential information.
- Information can be used or sold for various purposes such as when someone is stealing proprietary information of an organization, like research and development data.
- Data loss and manipulation is breaking into a computer to destroy or alter data records.
- An example of data loss is a threat actor sending a virus that reformats a computer hard drive.
- An example of data manipulation is breaking into a records system to change information, such as the price of an item

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- Identity theft is a form of information theft where personal information is stolen for the purpose of taking over the identity of someone.
- Using this information, a threat actor can obtain legal documents, apply for credit, and make unauthorized online purchases.
- Identity theft is a growing problem costing billions of dollars per year.

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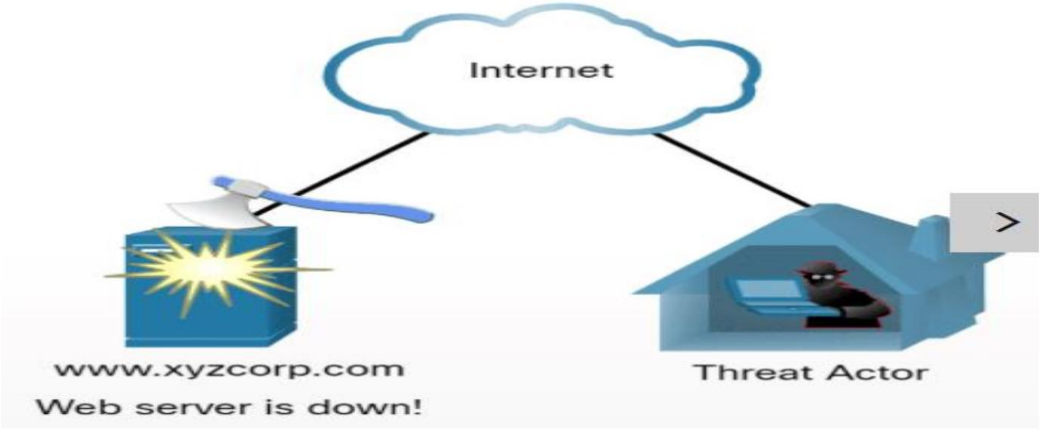
- Disruption of service is preventing legitimate users from accessing services to which they are entitled.
- Examples include denial of service (DoS) attacks on servers, network devices, or network communications links.
- Security threats from network intruders can come from both internal and external sources.

4.1. External Threats

- External threats arise from individuals working outside of an organization.
- They do not have authorized access to the computer systems or network.
- External attackers work their way into a network mainly from the internet through wireless links or dial-up access servers.



4.2. **Internal Threats**

- Internal threats occur when someone has authorized access to the network through a user account or has physical access to the network equipment.
- Internal attackers know the internal politics and people.
- They often know what information is both valuable and vulnerable, and how to get to it.

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- However, not all internal attacks are intentional.
- In some cases, an internal threat can come from a trustworthy employee who picks up a virus or security threat while outside the company and unknowingly brings it into the internal network.
- Most companies spend considerable resources defending against external attacks, however; some of the most damaging incidents are the result of actions by trusted internal users.

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- Lost smartphones and removable storage devices, misplaced or stolen laptops, and the failure to properly remove data from devices before disposal are common ways that user data ends up in the hands of the wrong people.

4.3. **Social Engineering attack**

- One of the easiest ways for an intruder to gain access, whether internal or external, is by exploiting human behavior.

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- One of the more common methods of exploiting human weaknesses is called social engineering.
- Social engineering is a term that refers to the ability of something or someone to influence the behavior of a person or group of people.

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- With these techniques, the attacker takes advantage of unsuspecting legitimate users to gain access to internal resources and private information, such as bank account numbers or passwords.
- Social engineering attacks exploit the fact that users are generally considered one of the weakest links in security.
- Social engineers can be internal or external to the organization, but most often do not come face-to-face with their victims.

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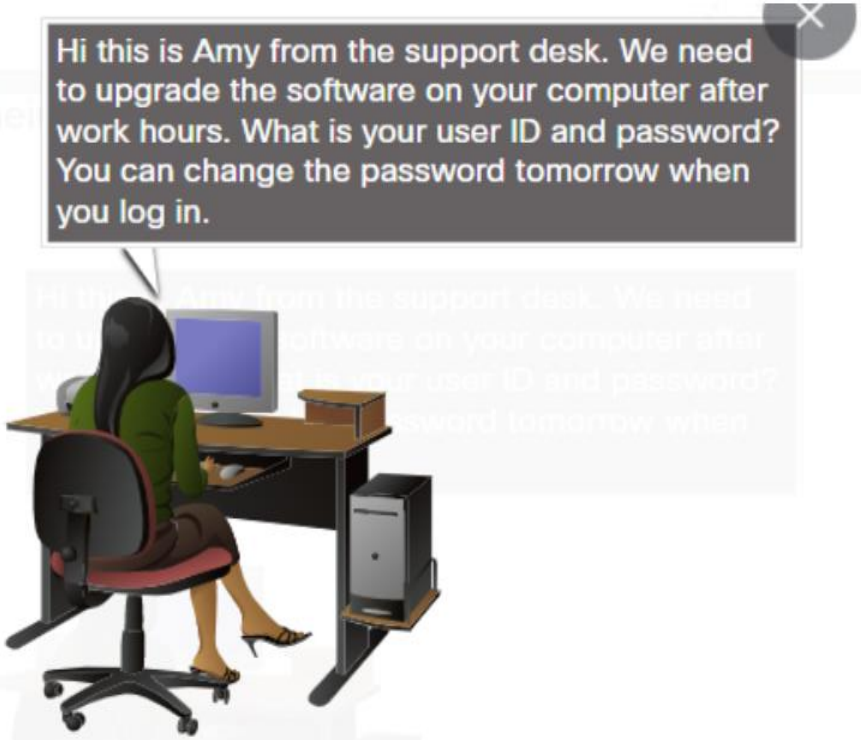
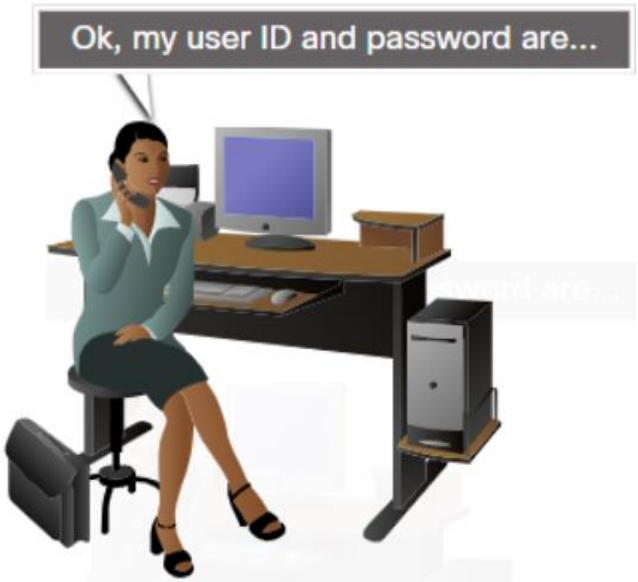


Social Engineer



Unsuspecting Employee
at Xyz Corporation

4.4. Malicious Software

- In addition to social engineering, there are other types of attacks launched by malicious software which exploit the vulnerabilities in computer software.
- Malware is the short name for malicious software.
- Examples of malware attacks include viruses, worms, and Trojan horses.
- All of these are types of malware introduced onto a host.
- They can damage a system, destroy data, as well as deny access to networks, systems, or services.

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- They can also forward data and personal details from unsuspecting PC users to criminals.
- In many cases, they can replicate themselves and spread to other hosts connected to the network.
- Imagine how difficult it would be to recreate saved files, such as game files, license key files, photographs, and videos.
- Sometimes these techniques are used in combination with social engineering to trick an unsuspecting user into executing the attack.

4.4.1. Types of Malware

- **Viruses** A virus is a program that spreads by modifying other programs or files.
- A virus cannot start by itself; it needs to be activated. When activated, a virus may do nothing more than replicate itself and spread.
- Though simple, even this type of virus is dangerous as it can quickly use all available memory and bring a system to a halt.
- A more serious virus may be programmed to delete or corrupt specific files before spreading.
- Viruses can be transmitted via email, downloaded files, and instant messages, or via CD or USB devices.

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- **Worms:** A worm is similar to a virus, but unlike a virus, it does not need to attach itself to an existing program.
- A worm uses the network to send copies of itself to any connected hosts.
- Worms can run independently and spread quickly. They do not necessarily require activation or human intervention.
- Self-spreading network worms can have a much greater impact than a single virus and can infect large parts of the internet quickly.

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- **Trojan Horses** A Trojan horse is a program that is written to appear like a legitimate program, when in fact it is an attack tool.
- It can not replicate itself. A Trojan horse relies upon its legitimate appearance to deceive the victim into initiating the program.
- It may be relatively harmless or may contain code that can damage the hard drive content of the computer. Trojans can also create a back door into a system that then allows threat actors to gain access.

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Spyware

- Not all attacks do damage or prevent legitimate users from having access to resources.
- Many threats are designed to collect information about users which can be used for advertising, marketing, and research purposes.
- These include spyware, tracking cookies, adware, and popups.
- While these may not damage a computer, they invade privacy and can be annoying.
- Spyware is any program that gathers personal information from your computer without your permission or knowledge.

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- This information is sent to advertisers or others on the internet and can include passwords and account numbers.
- Spyware is usually installed unknowingly when downloading a file, installing another program, or clicking a popup.
- It can slow down a computer and make changes to internal settings which creates more vulnerabilities for other threats.
- In addition, spyware can be very difficult to remove.

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Tracking Cookies

- Cookies are a form of spyware but are not always bad.
- They are used to record information about an internet user when the user visits websites.
- Cookies may be useful or desirable by allowing personalization and other time-saving techniques.
- Many websites require that cookies be enabled in order to allow the user to connect.

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Adware and Popups

- Three of the most common methods threat actors use to obtain information directly from authorized users go by unusual names: pretexting, phishing, and vishing.
- **Adware** is a form of spyware that is used to collect information about a user based on websites the user visits.
- That information is then used for targeted advertising. Adware is commonly installed by a user in exchange for a "free" product.

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- When a user opens a browser window, adware can start new browser instances which attempt to advertise products or services based on the surfing practices of a user.
- The unwanted browser windows can open repeatedly, and can make surfing the internet very difficult, especially with slow internet connections.
- Adware can be very difficult to uninstall.

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Popups and pop-unders are additional advertising windows that display when a website is visited.

- Unlike adware, popups and pop-unders are not intended to collect information about the user and are typically associated only with the website being visited.
  Popups - These open in front of the current browser window.
  Pop-unders - These open behind the current browser window.
- They can be annoying and usually advertise products or services that are not wanted by the user.

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Botnets and Zombies

- Another annoying by-product of our increasing reliance on electronic communications is unwanted bulk email.
- Sometimes merchants do not want to bother with targeted marketing.
- They want to send their email advertising to as many end users as possible hoping that someone is interested in their product or service.
- This widely distributed approach to marketing on the internet is called spam.
- One of the ways that spam can be sent is by using a botnet or bot.

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- "Bot" is derived from the word "robot" which describes how the devices act when they are infected.
- Malicious bot software infects a host, usually through an email or web page link, by downloading and installing a remote control function.
- When infected, the “zombie” computer contacts servers managed by the botnet creator.
- These servers act as a command and control (C&C) center for an entire network of compromised devices, which is called a botnet.

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- Infected machines can often pass the software to other unprotected devices in their network, increasing the size of the botnet.
- Some botnets include many thousands of infected devices.
- Bot software programs can also cause security issues on the infected machines.
- This is because the installed software may include the ability to log keystrokes, gather passwords, capture and analyze packets, gather financial information, launch DoS attacks, and relay spam.

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Denial of Service (DoS)

DoS attacks are aggressive attacks on an individual computer or groups of computers with the intent to deny services to intended users.

DoS attacks can target end-user systems, servers, routers, and network links.

DoS attacks are relatively simple and can be initiated by an unskilled threat actor.

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- A threat actor uses a DoS attack to perform these functions:

✓Flood a network, host, or application with traffic to prevent legitimate network traffic from flowing.

✓Disrupt connections between a client and server to prevent access to a service.

- There are several types of DoS attacks. Security administrators need to be aware of the types of DoS attacks that can occur and ensure that their networks are protected.

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Distributed Denial of Service (DDoS)

- DoS attacks that are coming from a single IP address can disrupt a website for a period of time until the attack can be isolated and defended against.
- More sophisticated types of attacks can bring web services offline for much longer lengths of time.
- DDoS is a more sophisticated and potentially damaging form of the DoS attack.
- It is designed to saturate and overwhelm network links with useless data.
- DDoS operates on a much larger scale than DoS attacks.
- Typically hundreds or thousands of attack points attempt to overwhelm a target simultaneously.

4.5. Security Practices and Procedures

- It is never wise to assume that your device or network will not be the next target of an attack.
- Taking protective measures can guard you from loss of sensitive or confidential data, and can protect your systems from being damaged or compromised.
- Security procedures can range from simple, inexpensive tasks such as maintaining up-to-date software releases, to complex implementations of firewalls and intrusion detection systems.

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- Some of the most effective security procedures are simple to implement and do not require extensive technical knowledge.
- A username and password are two pieces of information that a user needs to log on to a computer or application.
- When a threat actor knows one of these entries, the attacker needs only to crack or discover the other entry to gain access to the computer system.
- It is important to change the default username for accounts such as administrator or guest, because these default usernames are widely known.

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- Whenever possible, change the default usernames of all users on computers and network equipment.
- Most users select passwords that can be easily guessed or derived from known information about the user such as birthdays, pet names, or a favorite sports team.
- It is important to view passwords as a key to valuable data and to make them as secure as possible.

4.6. Security Tools and Applications

- Internet security is a major concern around the world.
- As a result, many tools are available to network users to protect the devices from attacks and to help remove malware from infected machines.

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Security Tool or Application	Description
Firewall	A security tool that controls traffic to and from a network.
Patches and updates	Software that is applied to an OS or application to correct a known security vulnerability or add functionality.
Virus protection	Antivirus software is installed on an end-user workstation or server to detect and remove viruses, worms, and Trojan horses from files and email.
Spyware protection	Antispyware software is installed on an end-user workstation to detect and remove spyware and adware.
Spam blocker	Software is installed on an end-user workstation or server to identify and remove unwanted emails.
Popup blocker	Software is installed on an end-user workstation to prevent popup and pop-under advertisement windows from displaying.

4.7. Signs of Infections

- Even when the OS and applications have all the current patches and updates, they may still be susceptible to attack.
- Any device that is connected to a network is susceptible to viruses, worms, and Trojan horses.
  - These may be used to corrupt OS code, affect computer performance, alter applications, and destroy data.

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So how do you know if your computer has been infected?

- Some of the signs that a virus, worm, or Trojan horse may be present include the following:

✓Computer starts acting abnormally

✓Program does not respond to mouse and keystrokes

✓Programs starting or shutting down on their own

✓Email program begins sending out large quantities of email

✓CPU usage is very high

✓There are unidentifiable processes or a large number of processes running

✓Computer slows down significantly or crashes, such as the when the Windows “blue screen of death” (BSoD) appears.

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- Antimalware includes a variety of software available to detect and prevent these types of intrusions and infections including antivirus software, antispam software, and antispyware software.

Antivirus Software

- Antivirus software can be used as both a preventive tool and as a reactive tool.
- It prevents infection. It detects and removes viruses, worms, and Trojan horses.
- Antivirus software should be installed on all computers connected to the network.
- Antivirus software relies on known “virus signatures” to find and prevent new viruses from infecting the computer.

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Some of the features that can be included in antivirus programs are:

- Email checking - Scans incoming and outgoing emails, and identifies spam and suspicious attachments.
- Resident dynamic scanning - Checks program files and documents when they are accessed.
- Scheduled scans - Virus scans can be scheduled to run at regular intervals and check specific drives or the entire computer.
- Automatic Updates - Checks for and downloads known virus characteristics and patterns.
- Can be scheduled to check for updates on a regular basis.

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Antispam Software

- No one likes opening their email and being overwhelmed by unwanted messages.
- Spam is not only annoying; it can overload email servers and potentially carry viruses and other security threats.
- Additionally, people who send spam may use links within the emails to take control of a host by planting code on it in the form of a virus or a Trojan horse.

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- Antispam software protects hosts by identifying spam and performing an action, such as placing
  it into a junk folder or deleting it.
- Spam filters can be loaded on individual devices, but can also be loaded on email servers.
- Antispam software does not recognize all spam, so it is important to open email carefully.

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Antispyware software

Spyware and adware can also cause virus-like symptoms.

- In addition to collecting unauthorized information, they can use important computer resources and affect performance.
- Antispyware software detects and deletes spyware applications, as well as prevents future installations from occurring.
- Many antispyware applications also include detection and deletion of cookies and adware. Some antivirus packages include antispyware functionality.

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Popup blocker

Popup blocking software can be installed to prevent popups and pop-unders.

- Many web browsers include a popup blocker feature by default.
- Note that some programs and web pages create necessary and desirable popups.
- Most popup blockers offer an override feature for this purpose.

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Additional Safeguards

One of the most common types of spam forwarded is a virus warning.

- Although some virus warnings sent via email are true, a large number of them are hoaxes and do not really exist.
- This type of spam can create problems because people warn others of the impending disaster and so flood the email system.
- In addition, network administrators may overreact and waste time investigating a problem that does not exist.
- Finally, many of these emails can actually contribute to the spread of viruses, worms, and Trojan horses.

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In addition to using spam blockers, other actions to prevent the spread of spam include the following:

- Apply OS and application updates when available.
- Run an antivirus program regularly and keep it up to date.
- Do not forward suspicious emails.
- Do not open email attachments, especially from people you do not know.
- Set up rules in your email to delete spam that bypass the antispam software.
- Identify sources of spam and report it to a network administrator so it can be blocked.
- Report incidents to the governmental agency that deals with abuse by spam.

Thank You!